ADVANCING DATA STRATEGY AND GOVERNANCE

Astha Malviya



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Introduction - Companies House

About Companies House:

Companies House is a government body which is responsible for dissolving and incorporating limited companies **UK's official registrar of companies**. It manages one of the largest open corporate registers in the world, maintaining the records of over 4.9 million businesses.

- **Company Formation**
- **Companies House Online Filing**
- **Government Gateway (Gov.uk One Login)**:
- **ForgeRock (Identity Management)**:
- **Confirmation Statements**:
- **Filing of Financial Statements (Annual Accounts)**:
- **Director Changes and Shareholder Updates**:

Involves submitting essential information like company name, directors, and registered address.

Registration can be completed **online or by post**, with varying fees depending on the method & company type. It is a **critical legal step** for anyone looking to operate a business in the UK as a legal entity.



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Current Organisational Strategy

Modernise operations	Strengthen business environment	Improve data integrity	Enhance fraud detection
Strengthening compliance and security	Optimising user experience	Empowering business growth	Maximising operational efficiency
Inspiring trust and confidence	Maximising economic value	Combating economic crime	Ensuring resource efficiency

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Companies House & Company Formation

UK's official registrar of companies:

- Companies House is the official government body that registers and oversees companies in the UK.
- Example: A startup like "CycleTech Ltd" must register with Companies House to legally operate in the UK.

Registers company details

- It records essential company information such as directors, shareholders, and addresses.
- Example: When registering "Brilliant Tutors Ltd", the founder must submit the names of all directors and the company's registered office address.

Legal registration process

- Company formation through Companies House is a mandatory legal step to establish a UK business entity.
- Example: Before trading, "VeganGrocer UK" must complete the formation process to gain its company number and legal status.

Online and paper-based

- Registrations can be completed online or via traditional paper forms, offering flexibility to applicants.
- Example: "SpeedApps Ltd" registered online for quicker processing, while "Vintage Books Co." chose the paper-based method.

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Current Data Strategy

Companies House relies on a **robust data strategy** to manage and provide access to business information. Their data governance framework ensures compliance, security, and data quality while supporting transparency and public trust. Key aspects include:

Open Data Initiative: Making company records accessible for public use.

Compliance with Regulations: Ensuring adherence to the Companies Act 2006 and GDPR.

Data Accuracy & Integrity: Implementing verification mechanisms for submitted records.

Technology Infrastructure: Using digital services to enhance data accessibility and management.

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Data Governance Today

Self-reported data:

Companies submit their own information with limited external verification.

Example: Directors and shareholders enter their details without supporting documents.

Minimal real-time verification:

Few automated checks are done at the point of submission.

Example: No instant flagging if a disqualified director is listed during registration.

Public register:

All submitted company information is published openly for public access.

Example: Anyone can search Companies House to view company filings and ownership.

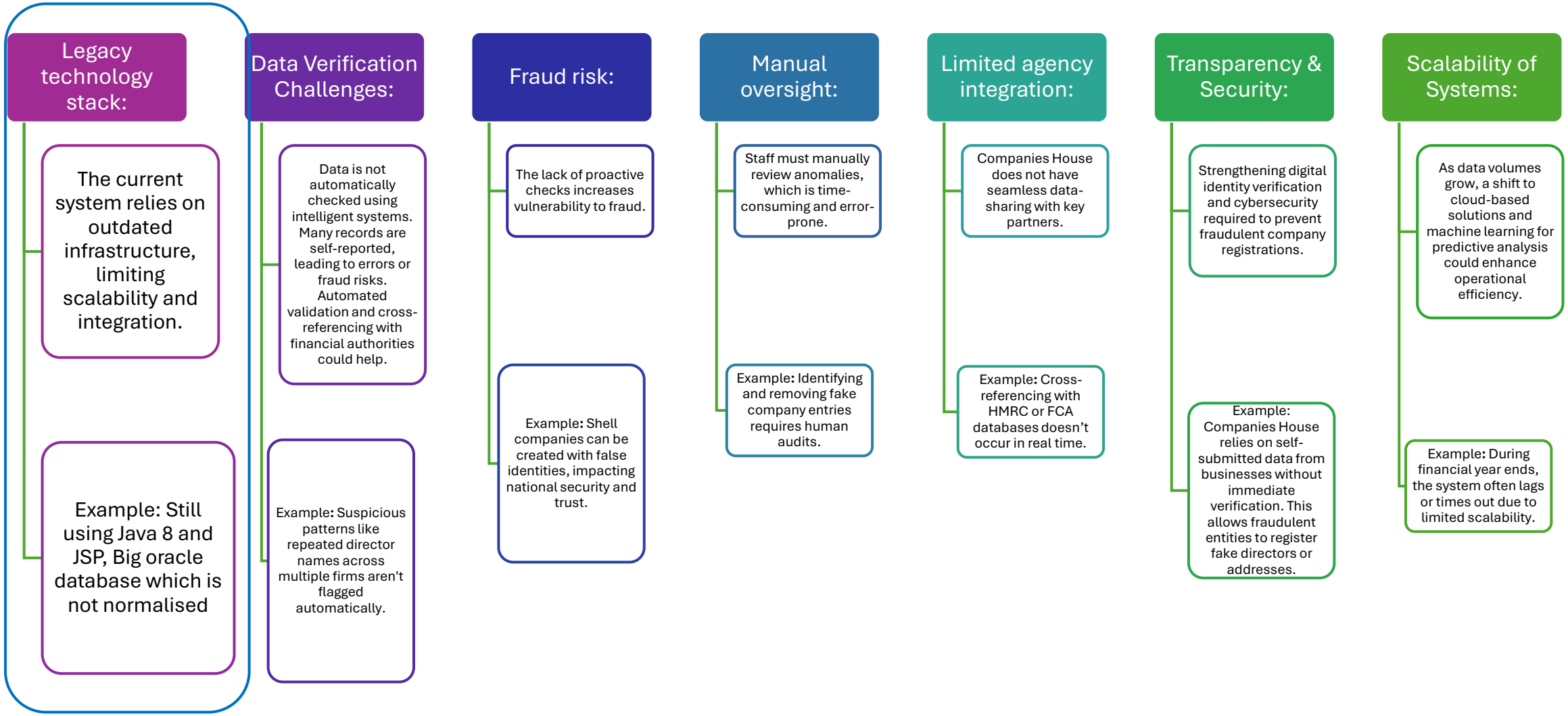
Reliant on honesty:

The system depends heavily on businesses submitting truthful data.

Example: False addresses or nominee directors may go undetected unless manually reviewed.

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Gaps in Current Strategy



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Proposed Change – Legacy Technology Stack Modernisation

Modernise	• Modernise core architecture : Move from Java 8 and JSP to cloud-native, microservice-based architecture (e.g., Spring Boot, Node.js, containerized with Docker/Kubernetes).
Adopt	• Adopt cloud infrastructure : Shift from on-premises to hybrid/public cloud (AWS/GCP/Azure) for scalability and disaster recovery.
Enable	• Enable API-first approach : Build secure RESTful APIs for real-time integration with HMRC, FCA, and others.
Upgrade	• Upgrade frontend : Move from static, JSP-based frontend to dynamic, accessible UI (React/Angular) with mobile responsiveness.
Fix	• Fix Legacy Database Bottlenecks: massive database holds millions of records but suffers from poor performance due to insufficient indexing and lack of normalisation.
Embed	• Security-first design : Embed modern authentication (OAuth2), logging, and compliance (GDPR, NCSC standards).

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Strategic Alignment

Why this investment matters:

Efficiency & Speed:
Legacy systems cause long deployment cycles and downtime.

Cost savings over time:
Maintenance of legacy systems is expensive and unsustainable.

Cybersecurity:
Older frameworks lack support and pose security risks.

Developer productivity:
Modern stacks attract and retain top tech talent.



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How the proposal aligns with Companies House's broader goals:

Supports Digital Strategy 2020–2025 :

Legacy systems cause long deployment cycles and downtime.

Boosts trust in the register :

Maintenance of legacy systems is expensive and unsustainable.

Future-ready architecture :

Older frameworks lack support and pose security risks.

Improves stakeholder experience :

Modern stacks attract and retain top tech talent.



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Change Management Approach

Kotter's 8-Step Change Model

Step	Action	Objective
Create Urgency	Highlight fraud and inefficiency risks	1. Publish internal whitepaper on risks of legacy systems 2. Stakeholder workshops 3. Impact assessments
Build Coalition	Engage leadership, IT, legal, policy units	1. Form cross-functional steering committee 2. Identify change champions 3. Leadership engagement sessions
Form a Strategic Vision	AI-based validation to support data integrity	1. Define future-state architecture 2. Choose stack 3. Document APIs and dev roadmap
Communicate Vision	Internal workshops, stakeholder outreach	1. Town halls 2. Intranet blogs 3. Dev showcases 4. Internal dev conference, external stakeholder Q&A
Remove Barriers	Infrastructure, training, pilot access	1. Upgrade DevOps tools 2. Build API gateway 3. Migrate off JSP 4. Launch pilot registry module with new tech stack
Generate Short-Term Wins	6-month AI pilot project	1. Deliver first new registry microservice 2. Deploy 2 MVP features in live environment 3. Gather stakeholder feedback
Sustain Acceleration	Expand and refine post-pilot	1. Extend modernised modules to cover all registration workflows 2. Integrate machine learning for data verification 3. Scale user load testing
Anchor in Culture	Make AI part of business-as-usual	1. Revise org charts 2. Create permanent agile squads 3. Annual review of performance KPIs 4. Start planning next 5-year tech strategy

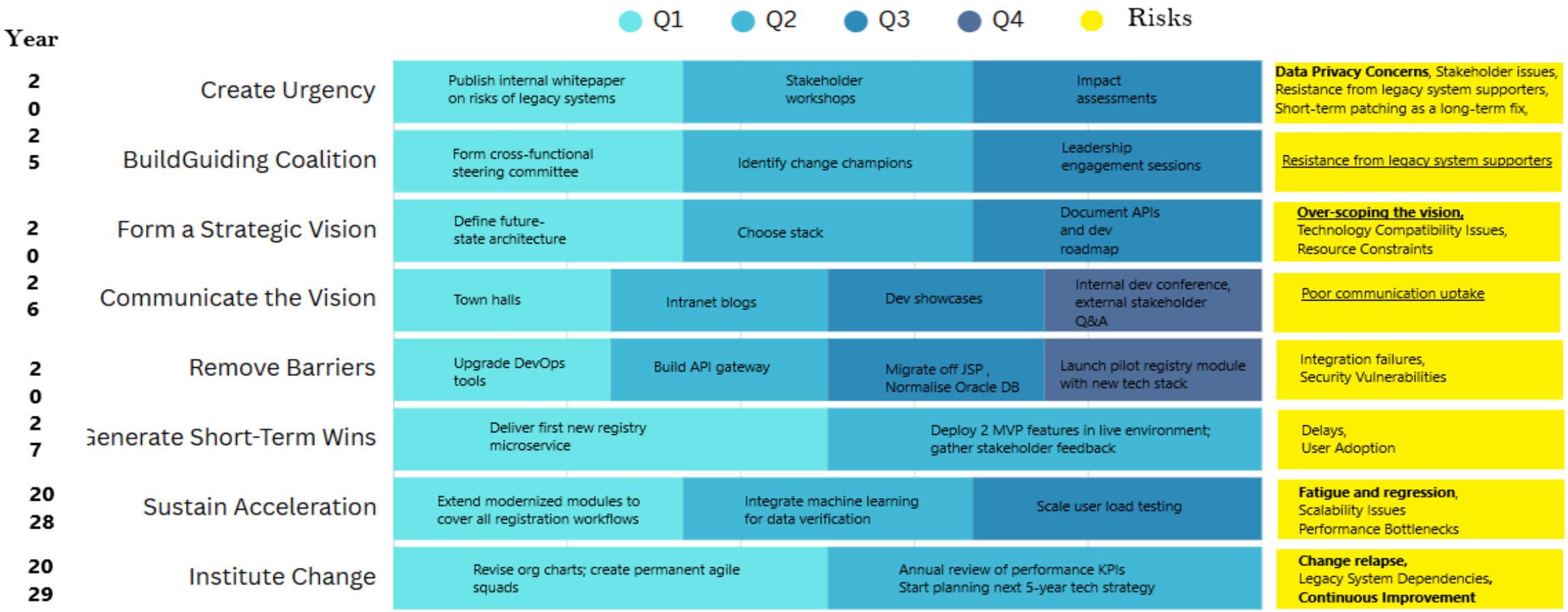
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Risk Mitigation & Scalability

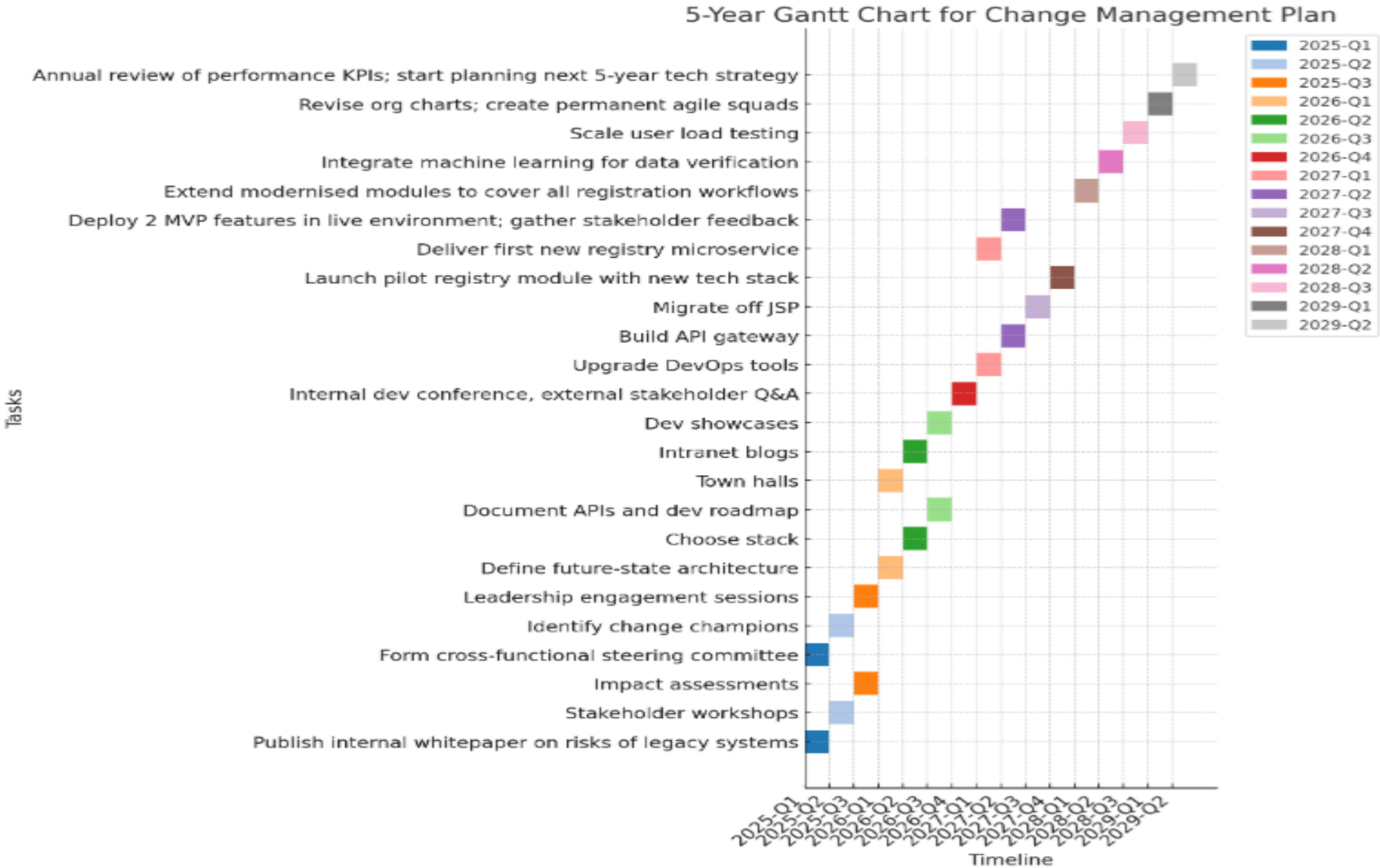
Risk	Type	Likelihood	Impact	Mitigation
Stakeholder Misalignment	Cultural	Medium	High	Early stakeholder involvement & persuasive data in whitepaper
Data Privacy Concerns	Communication	High	Medium	Clear messaging in workshops and assessments
Resistance from legacy system supporters	Organizational	Medium	High	Diversity across departments and levels
Change champions lack authority	Structural	High	Medium	Ensure exec backing and visible leadership support
Over-scoping the vision	Strategic	Medium	High	Engage architects early in the planning
Technology Compatibility Issues	Technical	Medium	Medium	Regular tech trend reviews; flexibility in design
Poor communication uptake	Communication	Medium	Medium	Use multi-channel strategies and feedback loops
Vision gets diluted over time	Cultural	Low	High	Reinforce messaging through leaders & champions
Security Vulnerabilities	Technical	High	High	Phased rollout with rollback options
Integration failures	Technical	High	High	Parallel systems
API gateway delays due to integration	Technical	Medium	High	Early testing with simulated environments
MVPs do not meet user expectations	Product	Medium	Medium	User feedback loops; agile iteration
Stakeholder fatigue from small wins	Cultural	Medium	Medium	Celebrate wins visibly and tie to broader vision
ML integration complexity	Technical	Medium	High	Pilot ML models; partner with data experts
Scalability issues during testing	Infrastructure	Medium	High	Cloud-based test environments; performance benchmarking
Agile squads lack continuity	Organizational	Medium	Medium	Clear role definition and career paths
Strategic planning neglects lessons from past 5 years	Governance	Medium	High	Post-mortem reviews; knowledge management systems

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5-year Visualisation



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DEI- Smart Objectives

Objective	Specific	Measurable	Achievable	Relevant	Time-bound
Increase Representation of Underrepresented Groups	Representation should increase for women and minorities in leadership roles.	Achieve a 20% increase in the number of minorities and women.	Implement targeted recruitment strategies and leadership development programs.	Company's commitment to diversity and inclusion should be Aligned.	Achieve this goal by the end of 2025.
Enhance Employee Awareness and Training	Provide DEI training to all employees.	Ensure 100% of employees complete the DEI training program.	Develop and roll out an online DEI training module.	Supports the company's goal of fostering an inclusive workplace culture.	Complete the training rollout by Q3 2025.
Improve Employee Engagement and Inclusion	**Increase the inclusion score in the annual employee engagement survey.**	**Achieve a 15% increase in the inclusion score.**	**Launch employee resource groups and regular inclusion workshops.**	**Enhances employee satisfaction and retention.**	**Achieve this goal by the end of 2026.**
Promote Pay Equity	Conduct a pay equity audit and address any disparities.	Ensure 100% pay equity across all roles and levels.	Partner with an external consultant to conduct the audit and implement recommendations.	Ensures fair compensation practices.	Complete the audit and adjustments by Q2 2026.
Increase Supplier Diversity	Increase the number of contracts with diverse suppliers.	Achieve a 25% increase in contracts with women-owned, minority-owned, and other diverse suppliers.	Develop a supplier diversity program and outreach initiatives.	Supports the company's commitment to economic inclusion.	Achieve this goal by the end of 2027.

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How this effects my role as a developer and Lead



Directly addresses pain points:
Targets the root issues of fragility and poor maintainability in the current legacy system.

Improves system stability:
Modern architecture and technology stack will reduce errors and system crashes.

Enhances developer experience:
Clean, modular code and modern tools will attract and retain skilled developers.

Boosts productivity:
Reduces time spent on debugging legacy issues and allows focus on feature delivery.

Supports agile delivery:
Enables iterative development and quicker turnaround of updates and fixes.

Facilitates onboarding:
Easier for new team members to understand and contribute to a well-documented, standardised codebase.

Builds team morale:
Working on modern systems fosters a sense of progress, pride, and engagement among developers.

Future-proofs development:
Aligns tech stack with industry trends, ensuring long-term relevance and compatibility.

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ROI & Business Impact of Effective Data Strategies

Return on Investment

Reduced Operational Costs:
- Automation and AI reduce manual reviews, saving thousands of staff-hours annually.

Faster Time-to-Resolution:
- Real-time data validation shortens the verification lifecycle from days to minutes.

Fraud Loss Prevention:
- Early anomaly detection prevents fraudulent registrations and misreporting.

Improved Developer Productivity:
- Clean, indexed, and validated data makes APIs and systems easier to maintain and integrate.

Strategic Business Impact

Stronger Regulatory Compliance:
- Automated checks aligned with legal mandates increase trust from oversight bodies (FCA, HMRC, ICO).

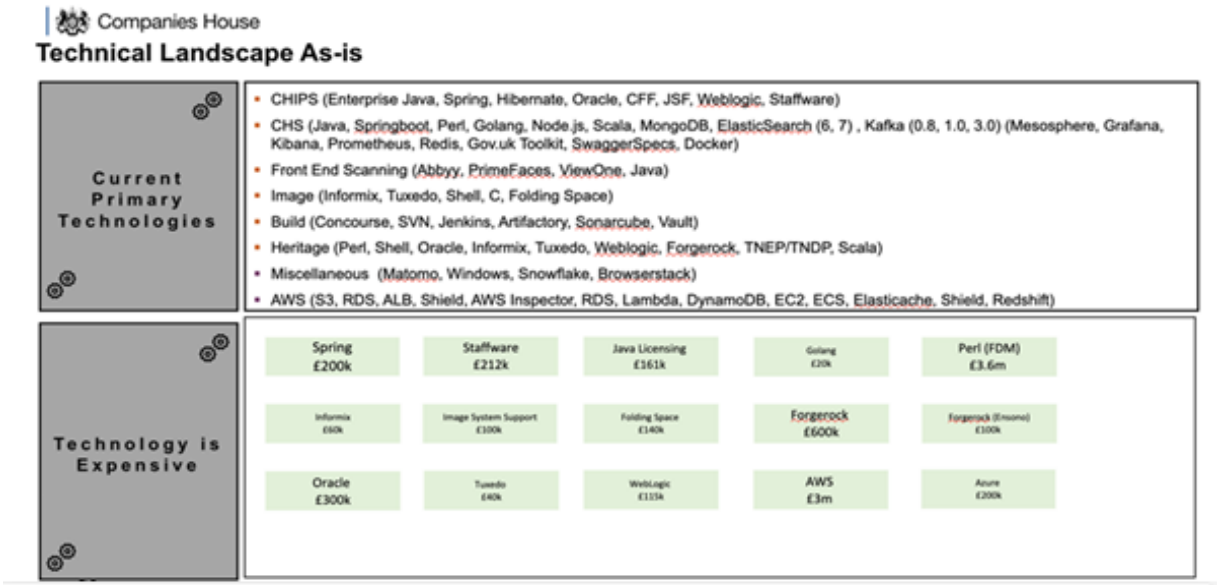
Boosted Public Trust:
- Transparent, accurate data builds credibility with citizens and businesses using the register.

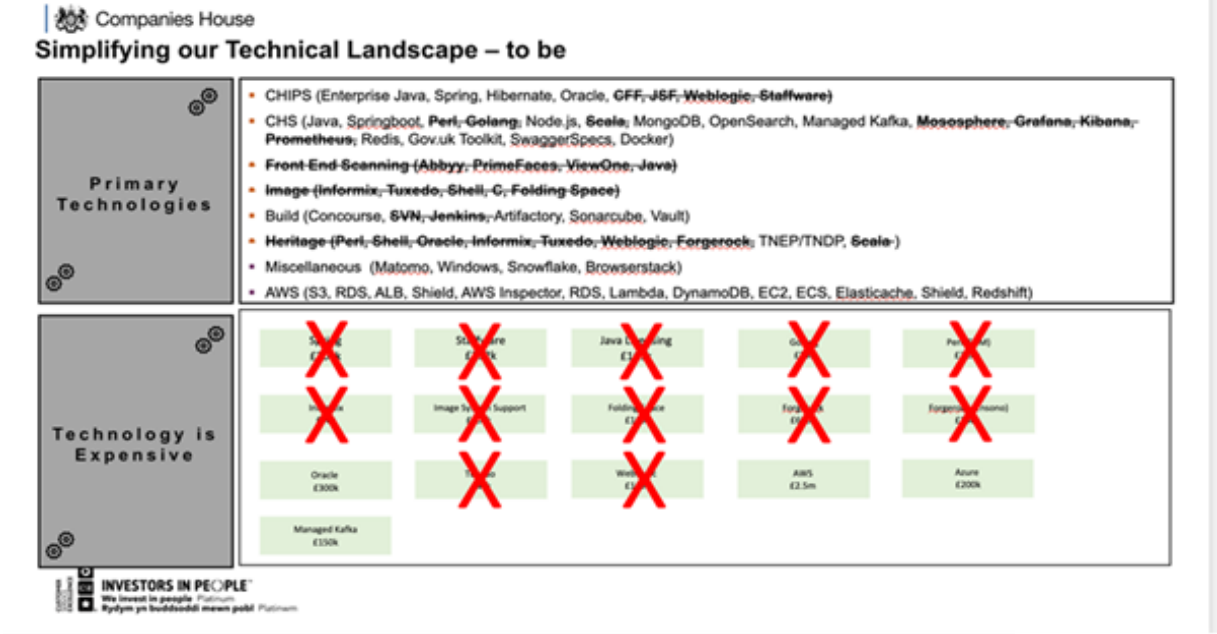
Scalability for Future Growth:
- AI-ready infrastructure supports scaling to millions of records without degrading performance.

Data as a Strategic Asset:
- Structured, normalized, and indexed data opens doors for analytics, forecasting, and policy shaping.

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ROI & Business Impact of Effective Data Strategies





INVESTORS IN PEOPLE
We invest in people Platinum
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- **ROI & Business Impact of Effective Data Strategies**

Benefit Area	Estimated Impact
Manual effort reduction	30–50% cut in workload
Error rate reduction	Up to 70%
Fraud prevention ROI	£200k+ per incident
Developer cycle improvement	~40% increase in speed

Staying Future-Ready

AI-Powered Anomaly Detection	Use machine learning to flag suspicious activities in company filings (e.g., sudden changes in ownership or address).
Natural Language Processing (NLP)	Automatically extract and validate named entities from documents, reducing manual input and errors.
Real-Time Cross-Verification	Integrate with external agencies (e.g., HMRC, FCA) to validate business data in real-time.
Fraud Mitigation through Pattern Recognition	Use AI to detect duplicate or shell companies based on historical data patterns.
Modular & Scalable Architecture	Future-proof system built to support plug-and-play integration of newer technologies like AI/ML services or Blockchain.
Transparency through Audit Trails	Design for future adoption of blockchain-based audit logs for tamper-proof historical records.

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Conclusion

Problem: Legacy stack limits Companies House's innovation and trust.

Solution: A 5-year modernisation roadmap using cloud, APIs, and secure microservices.

Outcome: Trusted, inclusive, scalable registry aligned with UK digital and economic goals.

Leadership Reflection: Kotter's model, ethical practices, DEI, and public value were central to our proposal.

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